

Information Security in Future Air Traffic Management Systems

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The detection, identification, and evaluation of vulnerable areas mark the cornerstone of the analysis that is required to improve the security of both current and future air traffic management systems. These three functions evolved from the use of new technologies in the communication, navigation, and surveillance infrastructure and in the new generation of aircraft. This analysis includes the identification of the assets, which are understood to be things of value or things to protect from potential threats. In the future context, an infrastructure for information sharing will be deployed to enable robust data exchange among air traffic management stakeholders. Information will become a major asset to be protected. This paper aims at presenting the future air traffic management environment by analyzing the impact of intended attacks on flight and surveillance information and how the effects are propagated throughout the whole system. For illustration purposes, some potential scenarios are described in order to facilitate the identification of applicable countermeasures, which can ensure confidentiality, integrity, and accessibility of the data.

I. Introduction

THE current air traffic management (ATM) infrastructure is reaching its capacity and efficiency limits. This is particularly true in Europe and the United States, where most congested sectors lead to flight delays, extra fuel consumption [often as a result of tactical air traffic control (ATC) interventions], and increasing environmental impact. Hence, a transformation of the ATM paradigm is needed to cope with the growing traffic demand at reduced operational costs and enhanced flight effectiveness while still improving safety and security.

Single European Sky ATM Research (SESAR) [1] in Europe and the Joint Planning and Development Office for the Next Generation Air Transportation System (hereafter referred to as NextGen) [2] in the United States have proposed initiatives to solve these problems. The plan is to evolve the current paradigm to the new trajectory-based operations (TBOs) concept, with advanced automated systems supporting humans in managing air traffic safely and efficiently. The TBOs concept is expected to bring substantial and tangible benefits to the global ATM system. The Federal Aviation Administration (FAA) has estimated that the application of TBOs will report a short-term cumulative benefit until 2018 equivalent to \$23 billion because of reduced time delays of 35% regarding current values. This would imply an associated reduction of fuel consumption of about 1.4 billion gallons, and therefore a decrease in CO₂ emissions of 14 million tons during the mentioned period [2]. Although these figures are promising, the major advantages of the new system are expected in the long term. The European Commission (EC) estimates that air traffic demand in Europe will grow to approximately 25 million commercial flights yearly by 2050, compared to 9.4 million in 2011. It has established ambitious performance targets for that year, with a maximum delay of 1 min per flight, a 75% reduction in CO₂ emissions per passenger and kilometer, and a 90% reduction in NO_x emissions, with less than one accident per 10 million flights (considering a heterogeneous traffic mix of manned and unmanned aircraft) [3].

The future ATM system will rely more on the knowledge and use of accurate and up-to-date information both on the ground and onboard. The system will be transformed from the current point-to-point infrastructure to a wide information-sharing infrastructure where all information is available to the stakeholders according to their needs, capabilities, and responsibilities. This infrastructure is called the systemwide information management (SWIM) infrastructure in both Europe and the United States. SWIM will facilitate the information exchange, providing a seamless infrastructure for leveraging all collaborative decision-making processes. In this new environment, the aircraft becomes one of the intelligent nodes that provides and consumes information to support such processes.

New onboard capabilities will improve the air–ground synchronization, enhancing the effectiveness of ATC directives thanks to the use of improved digital datalink communications, which are expected to reduce the miscommunications between flight crews and the air traffic controllers (ATCOs). Communication with Airline Operation Centers (AOC) will be also enhanced. Information technologies (ITs) used by airlines are being used more extensively as part of the maintenance, engineering, and flight operations. At the same time, airline manufacturers are designing and deploying onboard IT system architectures, which enable a substantial improvement in the technical dispatch reliability and a reduction in the maintenance and operations costs [4].

In this environment, information security is a critical enabler to implement the ATM systems of the future. ATM information will become one of the most important assets to be protected against unexpected or malicious attacks. The main vulnerabilities of the system may occur from the lack of adequate data protection according to their level of criticality. Different information packages will be considered more critical or less critical, depending on who generated them, when and why they were generated, and their availability and influence in further processes.

All these new system capabilities that are designed for evolving the current ATM infrastructure toward the SWIM concept will concurrently facilitate new alternatives for interfering with, disrupting, or diminishing the system performances, or even affecting the safety of operations. This paper exposes some threats that the authors believe need to be assessed in order to design countermeasures that reduce the impact of possible attacks based on the vulnerabilities of future systems. Descriptions of realistic scenarios are also included to evaluate the impact of individual actions on the system performances and how they propagate their effects throughout the whole system.

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The awareness for aviation security is not new. Aviation security has been present since the first releases of the International Civil Aviation Organization's (ICAO's) Annex 17 for security [5]. The applicability was limited to aircraft operators and airports, focusing primarily on hijacking and bomb threats. The well-known events of Lockerbie and September 11th triggered the growing concern for ATM security.[†] New awareness arose with the possibility of other types of security-related events, including attacks on air traffic service facilities and the communication, navigation, and surveillance (CNS) infrastructure. ICAO document 9854 [6] presents a vision for the integrated, harmonized, and globally interoperable ATM system. A key concept of this vision is the continuity of service. To maintain that concept, the need is to not only protect the ATM system against security threats but also to secure ATM operations. The ICAO's ATM security manual ([7] document 9985) addresses both the protection of the ATM system infrastructure and ATM security operations. It complements the Aviation Security Manual ([8] document 8973) and is expected to be the guidance material for countries to implement ATM security provisions.

Air transport stakeholders such as EUROCONTROL, the European Civil Aviation Conference, the ICAO, the Federal Aviation Administration, and the EC have joined forces to implement the guidelines and regulations needed to further improve ATM security worldwide. There is common agreement on the need to first assess in depth any possible attack to the infrastructure or malicious intervention (act of unlawful interference) from any intentioned individual or group. Such an assessment would be used for supporting all existing and new ATM operational concepts. In 2012, the ICAO published the ICAO Global Risk Context Statement [9] to help and guide global aviation leaders in taking a risk-based approach to maintain civil aviation security programs.

In Europe, EUROCONTROL published the ATM Security Risk Assessment Methodology [10] to provide the air navigation service providers (ANSPs) with a systematic approach to security risk assessment that is compliant with EC regulation 2096/2005 [11].

SESAR has also planned the development of an ATM risk assessment under SESAR Work Package 16.02.03—Security Management.^{***} Its main goal is to identify and evaluate sensitive assets, characterize threats targeting those assets, and evaluate the risk of associated risk scenarios. The North Atlantic Treaty Organization (NATO)/EUROCONTROL ATM Security Coordination Group (NEASCOG) is an international and cooperative initiative. It coordinates all stakeholders that have roles to play with respect to ATM security. This organization, which manages civil and military common interests, is focused on the description of high-level ATM security aspects, measures to improve ATM security, human resources awareness and cooperation, and the development of an ATM security assessment and monitoring methodology. This methodology is described in the NEASCOG Airspace Security Threat and Risk Assessment, a document regularly reviewed,^{††} which includes recommended or updated practices related to threat assessment methods.

All of these documents contain different ways to perform threat and risk assessments. Initially, the most common threats were usually physical attacks to the aircraft or to assets in the airport terminal. We now recognize the role played by information and communication technologies and the need to consider the vulnerabilities of a new ATM system that is very much based on information exchange. The detection, identification, and evaluation of areas of vulnerability in the technologies associated with CNS systems represent the core of a threat analysis that would contribute to improving security in the future ATM context. In this context, a common information exchange infrastructure would support the sharing of key data among stakeholders. Additional features of the e-enabled aircraft will also have to be considered because they may add new vulnerabilities to the system. For instance, the instantaneous network connectivity between aircraft and airline to support maintenance and operational data and Internet connection for passengers based on commercial wireless network datalinks will require an exhaustive analysis of potential vulnerabilities [12].

The next section summarizes the main challenges in transforming the current ATM system into the new one, with a special focus on the CNS elements that enable the capabilities of the new operational concept. Additionally, the recent e-enabled aircraft concept and the information channels with passengers and airline operation are also considered. This section also includes a brief description of the most relevant assets that will be the core of the threat assessment: the aircraft, the netcentric infrastructure, and the information.

Section III exposes the role of information in the future ATM context, by analyzing how this information can be modeled according to its nature, and the main characteristics of each information domain.

Section IV presents some vulnerabilities of the future ATM system, with special attention to the problem of scalable and multithreaded attacks. In most situations, attacks are not designed individually. The analysis of how multiple attacks to the same asset or individual attacks to different assets can affect the standard procedures is also studied. This section also collects relevant functionalities that need to be addressed by any implementation of the SWIM, taking into consideration the protection of the ATM information under any circumstance. Aspects of security programs established in other domains are also discussed in this section.

Section V describes potential attack scenarios where information is corrupted, manipulated, or disabled. It describes the effects on the whole ATM system and how operational safety is affected in each case.

Finally, Sec. VI gathers the main outputs and conclusions of the presented work.

II. Future ATM System

TBOs represent a huge shift from clearance-based to trajectory-based controls, where the reference business trajectory (RBT) is agreed upon in advance between the airspace users, the ANSPs, and the airports. The RBT is owned by the airspace users, whereas the ANSPs and airports are committed to facilitate the execution of the trajectory as long as safety is not jeopardized. This new ATM paradigm is characterized by a delegated aircraft-to-aircraft separation assurance and by the capability of providing the means for executing the RBT and the ability of following such a trajectory without major discrepancies [13,14].

With such high expectations, and considering trajectory management (TM) as the key feature of the envisioned ATM system, the technological challenges to be solved are related to how we can efficiently and securely connect all advanced systems (both onboard and on the ground) that are required to support such an operational concept.

This global connectivity will rely on an infrastructure to manage the information sharing. This infrastructure is based on a platform that interconnects all systems, thereby enabling robust and reliable information sharing. The infrastructure is called systemwide information management in both Europe and the United States. This netcentric concept considers all systems, including the aircraft, as network nodes that produce and consume information according to their respective needs. This information not only includes surveillance data (actual aircraft position and velocity) but also additional data related to weather forecasts, traffic flow, or trajectory predictions that facilitate the main ATC objective, thereby ensuring safe operations. It is important to realize that SWIM supports air-ground and ground-ground data exchange, as well as all ATM services; whereas air-air exchange is considered as outside the scope.

[†]Available online at http://en.wikipedia.org/wiki/Pan_Am_Flight_103 and http://en.wikipedia.org/wiki/September_11_attacks.

^{**}Available online at <https://www.atmmasterplan.eu/projects/16585>.

^{††}This document is not publicly available.

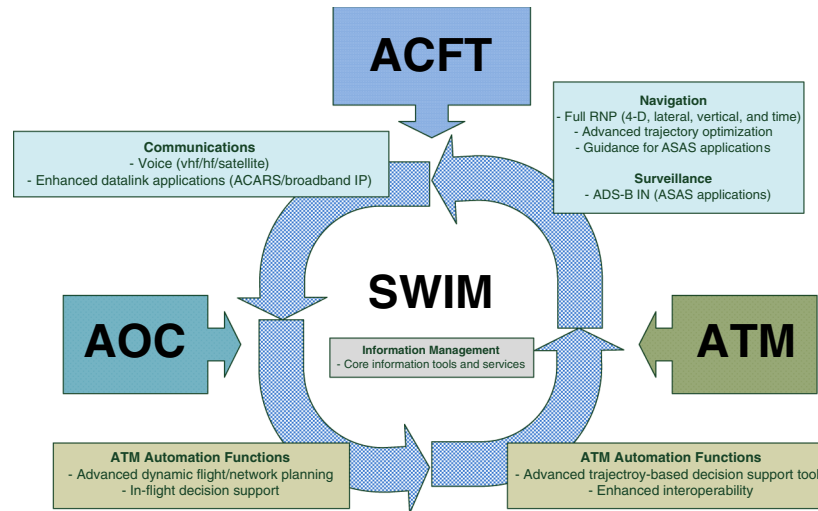


Fig. 1 Overview of future ATM system (ACFT, aircraft; IP, internet protocol; ACARS, aircraft communications addressing and reporting system).

Added to the deployment of the new ATM network, CNS improvements are paramount for achieving the expected operational targets. Better capability to communicate between air and ground, enhanced capabilities for aircraft positioning and guidance, and more precise traffic monitoring as a result of accurate surveillance information are the high-level targets to be reached.^{††} Figure 1 details the interdependencies among main elements of the future ATM system. To be specific, the elements are as follows:

1) The first element is communications. A future ATM concept demands datalink services supporting features such as four-dimensional (4-D) trajectory management, airborne separation assurance system (ASAS) applications, or information management. A reliable and efficient communication infrastructure will have to serve all airspace users in all types of airspace and flight phases, providing the appropriate quality of service needed by the most demanding applications. Basically, three different means of communication will be available to users: a) ground-based line-of-sight datalink for continental operations, b) satellite-based communications for oceanic operations while complementing the ground-based continental datalink as a way of improving the total availability, and c) airport dedicated links for sharing a significant amount of information with the aircraft, such as maintenance information or flight database updating.

2) The second element is navigation. Developments are predominantly focused on the evolution of the Global Navigation Satellite System-based navigation technologies expected for supporting the required navigation performance (RNP) operations. Moreover, the gained improvements about knowledge of actual positioning will conduct to a delegation of certain self-separation tasks from ATCO to flight crews. This will suppose that each aircraft will have to ensure minimum separation with surrounding traffic, whereas ATCO will act as supervisor, only intervening in critical circumstances.

3) The third element is surveillance. The extensive use of automatic dependant surveillance–broadcast (ADS-B) and wide-area multilateration (WAM) will deliver improved performances in terms of accuracy, information update rate, coverage, and efficiency regarding the primary surveillance radar (PSR) and secondary surveillance radar (SSR). Advanced multisensor surveillance data fusion will also contribute to optimize the use of common airborne components, depending on specific ground system requirements and the operational needs. Based on how the future surveillance is currently conceived, it is worth mentioning here that redundancy (i.e., multiple-layer surveillance based on radar, WAM, and ADS-B) presents the best security measure, both short term and medium term.

Additionally, wireless networks (satellite and terrestrial datalinks) will provide a seamless connection between aircraft and ground-based infrastructures [basically, ATC systems and airline operation centers (AOCs)], as well as the Internet-based passenger entertainment.

Because of these new CNS capabilities and onboard/offboard connectivity, the ATM infrastructure will provide more efficient processes and procedures to manage the air traffic in a safe manner where all stakeholders are interconnected with a more robust and accurate data exchange and broader capabilities to meet the user's preferences.

In the context of future operational context, it is very useful to organize the assets attending to the target of the operational improvements, which improve the system performances. This paper proposes three main domains that can be identified according to that assumption: aircraft, netcentric infrastructure, and information. This approach moves the focus of the assessment to the aircraft, which becomes the cornerstone of any analysis of system vulnerabilities. Current methods, such as the SESAR risk assessment methodology, basically consider the impact on ground-based infrastructure and how the impact on that environment can affect the traffic management. The study of this triangle of assets also includes the influences among them and the influences of the chain effects produced on the others by dysfunctions or disruptions in one of them.

This framework could make the design and development of future secure systems easier by the assessment of the impact on those three main pillars. For instance, the future datalink communications will require secure protocols to share the information among ATM stakeholders. A previous analysis of how those communications will affect the aircraft, the netcentric infrastructure or the actual information packages will help to create optimal designs, and therefore improved and secure systems.

A. Aircraft

The aircraft itself is the most important asset. Currently, there is a common view about how to protect an aircraft and secure its operations [15]. However, the enhanced interoperability with the ground-based infrastructure of future e-enabled aircraft [16,17] may provide new attack vectors.

Although the aircraft will be connected to the SWIM infrastructure for exchanging information with ATC/ATM systems, it will have other interfaces with enterprise networks (e.g., the airline maintenance network) or even direct interfaces with Internet proxies. The aircraft communications, network, and services are classified as aircraft control, airline information services, and passenger information and entertainment services [13]. This variety of connections opens the possibilities of unauthorized access to some systems, which could affect the standard aircraft operations.

^{††}Data available online at <http://www.sesarju.eu/programme/highlights/cns-communication-navigation-surveillance-technology-sesar> [retrieved 2014].

The concept of aircraft access to SWIM describes how the aircraft can be connected to the netcentric infrastructure during all flight phases and what services and data storage will be available for onboard systems. This approach considers the aircraft as an untrusted node that requires a virtual private network (VPN) connection to get access to authorized services and data. Hence, the air-ground connection must follow all security requirements established for ensuring a secure exchange of information, with special attention to the flight data downloaded from the aircraft that need to be in compliance with the network firewall requirements.

The risks, however, may come from other sources, represented by dedicated connections for software maintenance or updating, known as electronic distribution of software, or for offboard retrieval of airplane health monitoring data. As proposed by Robinson et al. [18], the threats can be categorized in two different sets according to the results of intended attacks:

1) The first category includes safety threats. The objective is to jeopardize the safety of the aircraft during flight. This objective may be achieved by corrupting some of the airplane-loadable software (coherent corruption) by a software misconfiguration (aircraft configuration does not match with software features), by a distribution of unsuitable software, or by blocking a critical software update that would solve a safety critical issue.

2) The second category includes business threats. The goal is to have an impact on the efficiency of common operations, provoking delays or unnecessary rerouting by late detection of useless software updates, by repudiation of necessary updates, or by generating false alarms due to simulated erroneous behavior of some system.

Future aircraft will enable the capability to connect to the Internet while airborne. This capability, highly valuable for the passengers, may offer additional gateways through which an adversary can get access to the onboard network if this network is connected to the onboard Internet network. Usually, onboard Internet access is provided as part of the passenger information and entertainment system. This system, whether or not it is connected to other onboard systems, must be specifically secured for obvious reasons. Any potential data exchange between this network and any other network onboard may represent a vulnerability that must be seriously taken into account to secure aircraft operations.

B. Netcentric Infrastructure

The current ATM system is evolving from point-to-point communication to a distributed netcentric service-oriented infrastructure where the information is available to stakeholders according to their needs. Security and cybersecurity are considered critical for future network-centric-based operations. Hence, it is paramount to analyze the security of the infrastructure supporting those operations from an industry perspective [19] in order to decide on the most efficient implementation of protective measures. This analysis should be based on four pillars: 1) identification of critical assets, 2) characterization of threats, 3) regulatory dependencies, and 4) definition of stakeholders (including the security solutions providers). Although this paper focuses on the first two, regulatory dependencies and stakeholders cannot be omitted in a complete security analysis.

The huge transformation of the current operation paradigm will rely on the development of the SWIM infrastructure. The system has been conceived as service-oriented architecture (SOA), which enables a seamless communication means among all stakeholders, providing timely and secure access to the information. This approach will elude the need of implementing ad hoc interfaces to communicate with each individual ATM system, and therefore will facilitate interoperability thanks to such common specifications. In return, the whole infrastructure will improve services efficiency and business operations. The SWIM concept requires, from a reliable message exchange, mechanisms that support the designed services. The NextGen program has defined the National Airspace System (NAS) Enterprise Messaging Services (NEMSs) as the standardized procedure for producing, modifying, and consuming data through all systems within the SWIM environment. This messaging infrastructure also provides information security management and a maintenance process for all SWIM services. The NEMS makes use of the FAA telecommunication infrastructure (FTI) assets as a means for enabling the transference of data among machines through the secure service access points. NEMS are conceived for supporting several communication protocols and data formats to ease data exchange among heterogeneous systems. The data transference is performed through the NEMS messaging node (NMN), which ensures the reliability and confidentiality of the communicated information. The NMN owns the routing logic required to find the appropriate path from origin to destinations according to the data content and the status of the network. It is also responsible for package rerouting in case of node failure.

NEMS are integrated with the NAS Enterprise Security Gateway (NESG), which ensures that the data exchange with systems outside the NAS is executed following the appropriate security policies. The NESG provides a wide range of security tools (passive or active) such as antivirus, firewalls, or VPN connectors. These elements are dedicated to protect the systems within the network from external intrusions separating both environments with high-level logics, which discriminate between authorized data exchange from known sources and unauthorized or unknown sources. The NESG differentiates two zones consisting of trusted and untrusted NMNs. Although trusted NMNs are allowed to interact directly with any of the stakeholders within the network, the external stakeholders get the information by accessing or subscribing to an untrusted NMN on the NESG via VPN connections to the FTI.

C. Information

In the future ATM environment, each actor will be expected to provide highly accurate information, especially flight data. Thus, information will become the key ATM asset to enable envisioned system functionalities. Different technological improvements are planned to ensure that onboard positioning can be calculated more precisely (extensive use of a GPS signal throughout the whole trajectory from takeoff to landing) and that there are enhanced information access capabilities (exploitation of the ADS-B protocol to improve onboard and onground situational awareness).

The future netcentric infrastructure will enable the capability of reliable and secure information exchange among all involved stakeholders. The secure and reliable delivery of the data is paramount for the success of next-generation ATM systems. The next section describes in detail what the role of information will be in the context of the future ATM system.

III. Role of Information in the New ATM System

The set of data required for seamless operations can be huge, and therefore has been organized in six different domains according to their nature and usage.

The SESAR information model, depicted in Fig. 2, gathers all data types in a common structure that will support the interoperable exchange of data. This model is founded on the ATM information reference model (AIRM) [20], which includes all relevant ATM information and metadata required to enable information management functions in an open and service-oriented environment. The AIRM provides a common reference for the operational language and information with the aim of providing semantic interoperability within the ATM system. The use of the AIRM as a common reference promotes ATM information standardization, not only in Europe but worldwide.

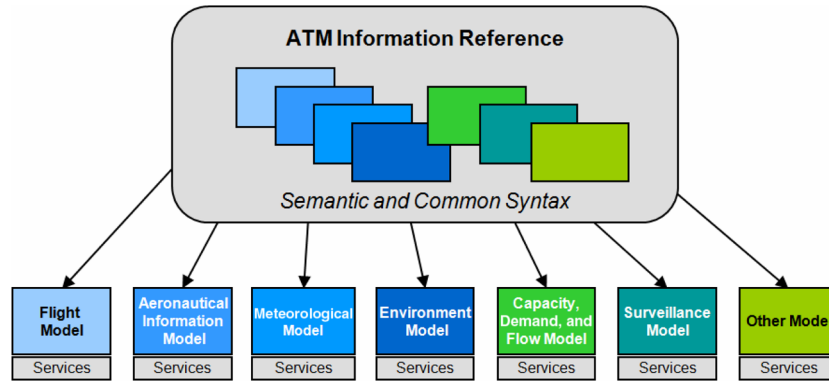


Fig. 2 SESAR information model [20].

The AIRM pursues the standardization of message content and format so that collaborating systems can have the same understanding of the ATM system state. Attending to the semantic rules and common syntax defined by such a metamodel, the ATM information that will be fed and distributed through the netcentric infrastructure is organized basically in six domains:

1) The flight domain includes all related information about a flight (aircraft identification, intended route, planned cruise speed and altitude, flight rules, equipment capabilities, etc.). The minimum information contained in (but not restricted to) an instance of this model is that included in the current ICAO flight plan. Additional data such as flight intent, aircraft intent, or 4-D trajectory data are examples of the type of information that is contained in this domain. The Flight Information Exchange Model (FIXM) [21] and the Flight Object Interoperability Proposed Standard [22] are compatible specifications for sharing flight information according to the rules imposed by the AIRM.

2) For the aeronautical domain, the provision of aeronautical information (airspace structure, waypoints and navigational aids, routes, standard procedures, aerodromes, etc.) will evolve from current aeronautical information service to the new aeronautical information management, which actually represents a transformation from product-oriented to datacentric operations. A single data source is required to provide such aeronautical data and can be accessible regularly or on demand in accordance with stakeholder needs. The aeronautical information exchange model (AIXM) [23] represents a formal structure for capturing all aeronautical information in a standardized format.

3) The meteorological domain is designed to enable meteorological information exchange among independent, harmonized, and interoperable platforms. This domain will facilitate a geo- and time-referenced access to meteorological information. This will improve both operational capabilities onboard by using accurate and updated wind predictions and atmosphere conditions, as well as ATC/ATM activities on the ground by exploiting a huge set of meteorological data. This will provide stakeholders with forecasts, observations, and measurements. It will alert them to hazardous weather phenomena and/or improve aircraft trajectory predictability. Similar to the FIXM and AIXM, EUROCONTROL and the FAA are promoting the use of the Weather Information Exchange Model.^{§§}

4) The environment domain gathers all applicable environmental regulations, which all ATM stakeholders need to respect. It delivers comprehensive and updated information about regulations, recommendations, and best practices. This model also includes a description of the main resources needed to meet the growing environmental challenge.

5) The capacity, demand, and flow domain was conceived for sharing information about the status of the airspace, its usage, and the foreseen incoming and outgoing traffic, as well as the imbalances between capacity and demand, which could generate undesired traffic congestions.

6) The surveillance domain is the standardized format for collecting surveillance information coming from different sources with the aim of tracking the traffic and watching safety rules. ATC tools will require a reliable and secure access to the data in order to control the traffic efficiently while ensuring separation minimum is not breached at any time.

All these domains and the related data models will facilitate the information management functions required for achieving the ATM system requirements. These requirements can be summarized as follows: enabling interoperability among heterogeneous automation tools and systems; open to be used by the community; platform independent; respecting global standards; and geospatially and time referenced.

From the perspective of the additional e-enabled aircraft features, the exchange of information between airline and aircraft helps airlines improve their business and enhance their operational efficiencies. The information architecture needs to be designed to enable this seamless transmission of data. Airlines can gather real-time information of the current fleet in flight by the exchange of the following data packages:

1) The first data package is maintenance. The airline operator can receive up-to-date fleet maintenance information. It also allows the distribution of technical information to the technical staff and the management of structural repairs and parts records.

2) The second data package is airplane health management. The information from the different sensors distributed within the aircraft allows real-time monitoring of the aircraft during flight. Once the aircraft is on the ground, the airline knows which spares are needed, whether any system has failed or has been damaged, or the maintenance status of each individual system.

IV. Vulnerabilities of the Future ATM System

Traffic surveillance activities in the future ATM environment will mainly rely on positioning, velocity, and timing information broadcast by the aircraft through the ADS-B system. This information is accessible to anyone having a receptor without any special restriction. The easy accessibility to such relevant data may facilitate exploitation by interested adversaries for disturbing and attacking indistinctly specific flights or the complete traffic within an airspace sector.

The main vulnerabilities of the CNS/ATM system can be categorized into three groups according to their impact: 1) spoofing or injection of fake flight information or fake ATC directives using a dedicated emitter [24]; 2) jamming or interference of some of the aeronautical frequencies used for surveillance or geolocalization [25]; and 3) cyberattacks, including the installation of malicious software (malware) in the ground-based infrastructure or in the onboard systems that might affect the adequate progress of a flight [26].

The study of how the situation might develop (including the reaction of the actors involved) represents a very useful approach to assess how a potential threat can affect the safety of the operations in a considered airspace.

^{§§}Data available online at <http://www.wxsm.aero> [retrieved 2014].

Knowing the existing security gaps and consequences helps in the implementation of the necessary countermeasure to ensure that the expected security levels are met. But, the implementation of countermeasures needs not only a previous knowledge of security gaps but also a sense for anticipating how the attack can be performed. The ways of impacting ATM systems are countless, especially because attacks tend to be more and more creative, new, and unique. One aspect to consider is the possibility of having potential, multiple attacks of the same nature executed at the same time (scalability) or multiple attacks of different natures executed following a predesigned plan and in a coordinated manner (multithreaded, similar to the multievent security scenarios tested during the FAA's network-enabled operations program [27]).

A single attack of those previously described might be managed relatively easily without generating a complicated situation (e.g., unintentional jamming near an airport can be solved without major impact in the airspace operations). But, when there are multiple attacks and attacks of the same nature are executed in different locations, the situation may become unmanageable.

On the other hand, the cybernature of some of the threats implies that an attack, like the injection of fake flight information in the ATM network, can escalate very fast, causing a chaotic situation when, not only one, but a huge number of ghost flights appears on controller consoles. This is what has been called a "scalable" attack. One suspicious flight can be confirmed by voice communication, but trying to filter out a high number of suspicious flights would overrun the ATCO capacity, leading to the collapse of the system if decision support tools (DSTs) are not able to discriminate real flights from ghost flights. As the attack escalates, the impact in the ATM system escalates too.

Although the analysis of the ATM system vulnerabilities requires a detailed study of the individual threats that can affect standard operations, it is well known that the design of intended attacks usually implies the coordination of different actions with a common objective. Each individual action is designed to achieve subgoals that facilitate the success of the global attack. Then, multithreaded attacks are considered as a set of coordinated actions, chronologically ordered, for which the main goal is to impact the performances of the ATM system, even jeopardizing the flight's safety.

The analysis of such attacks requires a general understanding, not only of the individual effects but of the ability of enabling new vulnerabilities. Some attacks could be considered as low-impact attacks when studied independently; however, they can be used by an adversary to distract the attention from the main attack thread, to act as a decoy, to facilitate the accessibility to later attacks, or to overload the deployed countermeasures.

At the same time, an adversary is highly unlikely to design simple attacks for hacking a very complex infrastructure. In the opinion of the authors, any unlawful act of interference against SWIM should be considered as the seed for more sophisticated attacks that can affect the stability of the system at different levels.

E-enabled aircraft provide the airlines with many benefits. However, they also present challenges from the security point of view. The information that manufacturers, airlines, aircraft, and aviation stakeholders share must be protected from cyberthreats; therefore, all computer systems, software, and offboard communication systems must be secured.

On the other hand, a new ATM paradigm from a security perspective represents a paradigm itself because some of the critical systems are "nonsecure-by-design" systems. To secure these systems, additional security measures and layers are required. This is neither efficient nor cost effective. The future evolution of ATM systems will require adopting the "secure-by-design" approach, as some air transport stakeholders (e.g., The Boeing Company) are already adopting. For example, the ADS-B system was intentionally designed as an open and public procedure for sharing flight information with the ground-based infrastructure and the surrounding traffic. However, it was not designed with any encryption alternative to protect the data against malicious use. The problem of integrating nonsecure-by-design systems into a global secured infrastructure is a huge challenge that may limit the capability of ensuring the expected overall system security.

A. Information Security in the Future ATM System

Information security represents the processes used for avoiding unauthorized access, use, disclosure, disruption, modification, inspection, recording, or destruction of a considered set of data [28]. The main goal of the information security is therefore to ensure that main information attributes are preserved throughout the whole data lifecycle. Those basic attributes are 1) confidentiality, which refers to the restriction of accessing privacy and proprietary information without explicit authorization from the owner; 2) integrity, which refers to the maintenance of accuracy and consistency of the information, avoiding unauthorized modification, deletion, or destruction; and 3) availability, which refers to the capability of accessing the information in a timely manner, and reliably, according to the needs of the user.

This list was enhanced by the Parkerian hexad [29], which proposes three new attributes to be considered in addition to the previous three: 1) possession or control, which refers to unauthorized changes of information ownership that could occur in breach of confidentiality; 2) authenticity, which refers to the veracity of the origin and authorship of the information; and 3) utility, which refers to the capability of using the information in accordance with the expected system features.

Those six attributes are used for establishing the security requirements to comply with any information-based infrastructure. This is the case of the future ATM environment, which will rely on sophisticated systems for sharing critical traffic information in a secure and efficient manner.

The future ATM data model will enable the release, update, and exchange of relevant information among the interested stakeholders according to the information management service model supported by the SWIM infrastructure. This mechanism for sharing information, which is the cornerstone of the envisioned system interoperability, is founded on a basic information management function that will extend from the generation to the distribution and exploitation of any ATM data. The most important functions considered for the development of the SWIM infrastructure are 1) quality, which describes the expected levels of accuracy, integrity, availability, resolution, or completeness of the information to be shared; 2) maintenance and updating, which describe the processes for maintaining and updating the data according to its relevance, criticality, or refreshing rate; 3) registration, which establishes the protocols for services publishing and subscription; 4) exchange and access, which declare the data formats defined for accessing the information or the mechanisms for exchanging the data in a robust manner; 5) user identity, which describes the privileges of each individual actor accessing the information according to the various contexts in which the user is registered; 6) security and authentication, which determines the levels of security for accessing the available information and the criteria for checking the authenticity of the actor requesting access; and 7) data ownership, which is the user's declaration of the legal rights to the information and the user's responsibilities on the reliability, accessibility, and accuracy of such data.

The main principles upon which the ATM information management procedures are designed imply the split between data provision and data usage. These principles are 1) georeferenced, which refers to data that need to be associated to a geolocation to facilitate the filtering process based on geographic positioning; 2) time referenced, which refers to data that need a time stamp to enable chronological sorting of relevant events; 3) seamless, which refers to the access to information and services needed to be global and universal according to the privileges of each individual stakeholder; and 4) open, which refers to the information and services needed to use standardized data containers, formats, and interfaces.

The development of the future ATM network will add new vulnerabilities to those of traditional distributed applications. For instance, service-oriented solutions commonly provide multiple Web services to external users, which leads to an increased unauthorized access susceptibility. In environments with highly limited external access, message-level security policies are practically unnecessary because all data generation,

transportation, and modifications are carried out internally. Message contents are assumed as valid because they have been processed by trusted systems without external influences. However, this is not the case of the future SWIM, where many services will exploit Web-based applications and whose strong requirement is the interoperability among heterogeneous systems. The main elements to take into consideration for evaluating the vulnerabilities of an SOA environment can be summarized in the following groups:

- 1) The first group is data format. The formal structure of messages, the type of information contained, and the policies followed for considering a message as well formed all represent a main source of system vulnerabilities. The use of standardized policy expressions for allowing interoperability among systems can also introduce some vulnerabilities.
- 2) The second group is services. These are system capabilities and features that determine the ability of the developed infrastructure to cope with the expected system functionalities and requirements. The implementation of such functionalities may allow third-party entities to get unauthorized access by the exploitation of services or Web interfaces, or through other elements.
- 3) The third group is service platforms. These elements provide messaging interfaces and communication infrastructure for supporting the network services.

A common awareness is required to avoid unnecessary risks. For example, it is widely accepted that most of the vulnerabilities come from the development of software applications. Therefore, following good practices such as input validation, security-oriented design, least-privilege approach, or sanitizing data to be sent to others can reduce the likelihood of a successful attack.

In addition, several security technologies can help to maintain the infrastructure as secure as possible. Basically, the applicable technologies are oriented to manage users or message content. The access control is the cornerstone for securing any SOA environment. It is critical to control who can get what. For such infrastructures where several different users (human or machine) with different levels of accessibility operate simultaneously, strong authentication is a must. Today, systems that use only a password-based authentication process can be easily hacked by brute force attacks. These attacks exploit the idea that the system strength is equivalent to the strength of the weakest password. Even the use of dynamic passwords has been demonstrated as an unreliable method for ensuring the robustness of access security. Hence, a two-factor authentication becomes the most realistic mechanism for reducing accessibility vulnerabilities. This type of solution usually requires a one-time passcode generating hardware tokens combined with a dynamic user password. Basically, this method needs something that the user has (e.g., a token generator or a badge) and something that the user knows (e.g., a code or a password).

From the security perspective, the content of the messages shared not only within the network but also with external entities needs to be monitored. For instance, the even more extended use of extensible markup language (XML) formats for sharing information among machines enables a new generation of attacks founded on certain common characteristics of those data packages as follows:

- 1) Denial of service consists of hampering or denying services by sending very large XML files that exhaust the machine capabilities, by sending thousands of XML messages that collapse the service, or by exploiting a recursive process that neutralizes the service.
- 2) Data integrity and confidentiality involves message tampering or sniffing techniques that allow access of the information contained in messages with the aim of modifying it or transmitting it to an outsider system. These processes can be used for modifying the requests of information to obtain additional data than what the service should originally return.
- 3) System compromise involves the service or the hosting server being disabled by the injection of malicious software embedded with the XML files or via stack overflow or buffer overrun.

The advantage of the SOA-based approach that is followed for designing and developing the SWIM infrastructure is that the vulnerabilities of such solutions are well known by the community; therefore, robust countermeasures can be directly used because they have been broadly tested in many other real infrastructures.

B. Establishing an Information Security Program

As stated earlier, information will become a main asset in the upcoming ATM paradigm. There are other sectors, such as the financial or the military ones, for which information has been traditionally considered as the business principal asset to protect, and where a series of processes and practices oriented to improve the information security level were long ago adopted and are now seen as essential (e.g., ciphering stored data and communications; applying a security dimension in every phase of the system lifecycle; and not relying only on security by obscurity, proprietary technologies, or closed systems as means to achieve a secure system). Now the aviation sector is striving to provide its critical information with a reasonable protection level. That information assurance is not a new issue; therefore, all lessons learned in other domains can be of great value for the aviation sector to avoid starting from zero and making the same mistakes in the implementation of a security structure.

Security measures regarding information security already implemented in other sectors that should also be adopted by the aviation sector may include [30,31] the following:

- 1) The first security measure is document security. During its useful life, information is subject to different and successive handling processes (e.g., transport, storage, custody, consultation, or destruction). A series of regulations should be applied in each of those processes. Measures regarding document security should include formal procedures to classify, register, distribute, store, and destroy documents, depending on the sensitivity of the information.
- 2) The second security measure is personnel security. This refers to measures that reduce the risk of staff exploiting their legitimate access to information for unauthorized purposes. Information security as a last resort depends on the people handling such information. Personnel security should not be considered resolved during the recruitment process. These measures include communication programs, awareness campaigns and formal processes for managing staff.
- 3) The third security measure is physical security. This refers to measures that prevent a direct assault or reduce the potential damage that can be inflicted in case of an incident of this kind. Physical security measures should also provide the necessary proof of any accesses or attempted accesses. Implementing a robust physical security model is critical for all businesses.
- 4) The fourth security measure is IT security. This refers to measures that protect information stored, processed, or transmitted by IT systems against the loss, whether intentional or not, of confidentiality, integrity, and availability of the systems handling such information.

The experience provided by other sectors teaches that information security should be addressed from a broad approach and not be limited to the acquisition and implementation of security technologies. To achieve successful results, and as a starting point, best practices recommend investing money, time, and manpower in defining, establishing, following, and maintaining a security program. An effective security program should include the following elements:

- 1) The first element is a security[¶] policy [32]. This is understood as the documentation of security decisions. A security policy is typically a document wherein specific requirements and rules that must be met for a specific area are established. It can be defined as a direction document for information security within an organization.

[¶]Data available online at <http://www.cni.es/> [retrieved 2014].

A complete policy should define the boundaries and goals of the security program, establishing the limits and resources (facilities, hardware, and software) covered by the program. It should also define a security management structure in order to clarify who has final responsibility for the security of the system. Therefore, clear responsibilities should be assigned to each involved actor.

An organization's security policy should address areas and topics of relevance for the organization (e.g., how to face the Internet connectivity issue). It is paramount to consider the security policy as a live element. The system to protect is continuously changing: new technologies are incorporated, whereas others become obsolete. Hence, the security policy cannot be defined once and remain static; it should be reviewed and updated as frequently as necessary.

To be efficient, it is recommended to focus the security policy on the decisions that should be made to protect a particular system and to be expressed as a set of rules. Policies should answer the question of who can do what and in which conditions.

Policies are written at a very high level, so they have to be complemented and supported by other tools such as standards, guidelines, and procedures to offer a clear view of implementation:

- a) Organizational standards declare the use of specific technologies or procedures that provide the benefit of the homogeneous use of common policies.
- b) Guidelines are focused in providing assistance to users in securing their systems. As systems may vary and evolve considerably, guidelines are developed to answer the question of what should be done but not how it should be done.
- c) Procedures are the most detailed documents. They are usually step-by-step guides to be followed in order to accomplish a specific task according to the security policy.

2) The second element is periodic risk assessment.^{***} To determine proper and useful policies, standards, guidelines, and procedures, it is basic to understand the events that may have a negative impact on the organization's performance. A thorough understanding of the possible risks can be achieved by following a formal risk assessment methodology.

Risk shows what could happen to the system assets if they were not properly protected and a threat was materialized. A risk value can be obtained by combining the likelihood of occurrence of a malicious event for the system and the consequences over the system performance. Absolute security does not exist; thus, it is important to establish a level of accepted risk. It will be determined by the quality threshold required by the system.

Risk analysis is a methodical approach to interpret risk. Independent of the methodology followed, there are always some common activities when carrying out a risk assessment:

- a) The first step is context establishment. The first step should be the identification of the system to be analyzed and what part or parts of the system will be considered in the analysis.
- b) The second step is assets identification. An asset is any resource with value for the organization that is necessary for the correct operation of the system. This task should include information, hardware, software, facilities, and personnel. Dependencies between assets have to be identified. One asset depends on another if the security needs of the first affect the security needs of the second. Assets should be valued in terms of possible damage if compromised.
- c) The third step is threat identification. Threats are events that may happen and cause damage to the target assets, harming the system. Threats are typically classified as natural disasters (e.g., floods, earthquakes), industrial accidents (e.g., electrical failures), and threats caused by people (both outsiders and insiders). Not all threats affect all assets, and a certain threat may not affect every dimension of a particular asset. Threats have to be valued in terms of degradation (the amount of damage caused) and frequency (how often the threat appears).
- d) The fourth step is impact determination. The objective of this step is to measure the damage caused by the materialization of a threat. This value can be easily calculated if assets have been correctly valued in all their dimensions and the degradation caused by the threats has been defined.
- e) The fifth step is risk determination. Finally, in this step, the risk (understood as the probable damage to the system) can be determined. Once the impact is known, the risk can be calculated by taking into account the likelihood or frequency of occurrence. The risk increases with the impact and with the frequency.

3) The third element is identity management [33]. It is very important to be able to identify without a doubt the different actors of the system, to have accountability, and to know who did what. Each actor should be assigned a digital identity. This identity will be managed during the whole lifecycle according to the current security requirements: issuance, expiration, renewal, revocation, and protection.

By establishing processes of identity management over a system, individuals are identified and it is possible to control their access to the system and to the resources of that system. This is done by defining links between entities, rights, and restrictions. For this purpose, there should be a clear definition of what users are allowed and are not allowed to do, as well as the specific tasks regarding each individual resource of the system. Identity management is closely related to the access control concept. By enforcing proper access control, it is well determined who can interact with what and under what conditions. It involves the establishment of a series of rules to allow or deny the access to certain users to certain resources and the access rights (e.g., data reading, writing, or deletion). A resource could be a specific system, some data, or even a certain facility.

4) The fourth element is computer security incident handling [34]. An incident is any violation of policy or law, or other unacceptable act, that involves any system information assets. It is highly recommended to cautiously prepare the planning and best practices so that the better response is provided when an incident is materialized.

An incident handling plan should include the following phases:

- a) The first phase is preparation. This is the most crucial phase. The ability of the organization in responding in an efficient way is determined by how well prepared the organization is. Preparation includes policies, response strategy, communication plan, access control, and tools.
- b) The second phase is identification. This refers to detection of the incident. This phase implies the monitoring of different sources to determine whether an incident has been or is being produced.
- c) The third phase is containment. Once an incident has been identified, the first goal should be the limitation of damage and prevention of further damage. Steps in this phase include short-term containment, backing up the system, and long-term containment.
- d) The fourth phase is eradication. In this phase, the affected systems are restored. It is crucial to be completely sure that malicious content has been removed and affected assets are completely clean before reincorporating them into the production environment.
- e) The fifth phase is recovery. In this phase, affected systems are reincorporated and the system recovers its normal functioning mode. It is key to confirm by monitoring that the system is behaving as expected.
- f) The sixth phase is lessons learned. By documenting all the processes undertaken during and after an incident, some lessons will certainly be learned. These lessons should be used to improve the preparation level, thereby closing the incident handling cycle.

5) The fifth element is training and awareness. It is equally important to promote security awareness and education among the users, who are often the weakest link in the security chain. Even the best security program can be compromised without the active collaboration of the

^{***}Data available online at <http://www.cpni.gov.uk/> [retrieved 2014].

people involved in the information systems. No matter how sophisticated the technology, security can always be circumvented via human intervention.

Efforts should be centered on building a security culture across management levels in a periodic fashion to achieve a proper understanding on cybersecurity and to ensure the proper support and funding. To settle this security culture, there are some basic cornerstones:

- a) Maintain and disseminate a clear and updated corporate security policy.
- b) Provide continuous training programs at all levels. They should include not only routine precautions and good practices but also specialized activities, depending on the role and responsibility of each work position.
- c) Security should be viewed as unobtrusive (it should not be an obstruction for the day-by-day activities). It should be natural and practiced by all management levels (as a good example to follow by all the staff).

Each of the elements described should be customized and developed for each actor involved in the ATM picture in order to develop personalized information security programs. An organization must view its security program as one of its most powerful security tools. Therefore, it should be thoroughly developed and continuously updated to be considered useful for the purpose of improving the information security level.

V. Characterization of Potential Attack Scenarios

Based on the vulnerabilities discussed earlier, this section describes possible situations or potential attack scenarios that help the community to understand potential ATM security threats and potential outcomes of such situations. This will help an organization design and develop a set of preventive strategies and countermeasures to avoid, or at least minimize, possible impacts.

ATM is ultimately the combination of technologies and the human to provide a safe guiding of aircraft in flight and on the ground. From the point of view of the scenarios described hereafter, information is considered the main asset to be protected, and therefore the potential target of any intended attack. Information is considered as any analogical or digital data shared between the aircraft, the ATC, the airline, or the ground infrastructure.

The examples provided in this paper focus on possible vulnerabilities of the technology that supports the flow of information among stakeholders, including the aircraft. The corruption, interception, jamming, or unauthorized injection of information will produce different consequences, depending on the flight phase, type of information affected, and the adversary's capabilities.

The scenarios are described as follows: 1) description of the operational scenario to be analyzed, 2) characterization of intended attack, 3) primary impact of the attack on target assets, and 4) propagation throughout the ATM system and secondary effects.

For all cases, it has been assumed that the final goal of the attack is to diminish the ATM performances, even affecting the operational safety, by designing attacks with the objective of corrupting ATM-related information or denying access to such information.

Most critical operations, where the traffic is more dense and complex, are the descents, final approaches, and landing phases. These operations are usually coordinated with departures from the same aerodrome and with crossing flights at cruise altitude. Hence, for the definition of the attack scenarios, the terminal maneuvering area (TMA) has been considered as the operational environment selected by the adversary to execute a planned attack. Table 1 shows the most relevant assumptions considered for describing a realistic environment in which the attack designed by an intended adversary will take place.

The examples of the following sections are considered as individual attacks. However, some coordinated actions are also presented to provide a more feasible description of a possible attack. More realistic scenarios should consider a wide variety of multithreaded attacks, which combine different ways for exploiting the system vulnerabilities with a common goal. Nevertheless, the objective of this paper is to present simple cases to be studied separately for further identification of appropriate countermeasures capable of protecting individual vulnerabilities.

A. Spoofing Scenario

In ATM, spoofing consists of the capability of supplanting the identity of a flight or several flights by injecting coherent information into the system. The result is the appearance of ghost aircraft, which do not exist but are detected by the surveillance infrastructure and displayed on the controller's console.

This effect can be achieved by the strategic location of an emitter of coherent ADS-B information in the vicinity of an airport. The signal will be received by the dedicated antennas and made available in SWIM by the corresponding system. This would generate a ghost flight that does not correspond to any real aircraft but is considered as real as long as data are not verified or filtered by other systems. To avoid filtering techniques, it would be necessary to file the related flight plan into SWIM; otherwise, there would be surveillance data for an aircraft without a designated flight plan. This is a required coordinated action that can be achieved by hacking the software infrastructure by means of a dedicated malware. For that purpose, the adversary needs to get access to the netcentric infrastructure, thereby exploiting a vulnerability of any of the system nodes.

Another means for injecting ghost aircraft is to use a malware with the capability of generating and exchanging flight data according to the expected formats and time intervals. This cyberattack requires the ability to execute a software piece without being detected but one that is designed to avoid any kind of filtering processes. Table 2 summarizes the described attack scenario.

In a worst-case scenario, it would be possible to have a huge number of ghost flights that could eventually collapse the whole system. In this situation, if filtering processes are not able to eliminate those flights, it would be impossible to separate the traffic in an efficient manner because the ATCO workload and DST capacity could be exceeded; therefore, reverting to procedural operations may be required to maintain acceptable safety levels.

Table 1 Operational scenario assumptions

Scenario	Description
Operational	Traffic descending to a congested aerodrome Surveillance based on the use of ADS-B information as primary system WAM and SSR/PSR as backup/redundant surveillance (no redundancy is also contemplated in some cases) No weather hazards are considered SWIM infrastructure ready for data exchange among ground-based systems Aircraft operating as a SWIM acting node Control pilot datalink communications as primary communication means between flight crew and air traffic controllers Advanced technology aircraft equipped with e-enabled features

Table 2 Description of a spoofing scenario

Scenario	Description
Intended attack	Introduction of fake information about a flight that is recognized by surveillance infrastructure as real and valid data Interfering nominal operations from descending flight to affect standard procedures Increasing the need of ATC interventions to preserve separation minima with unexpected traffic
Primary impact	System disruption by injecting a huge number of ghost flights that cannot be managed by either the ATCOs or by automation tools Increased ATCO workload due to managing a higher number of flights than expected Diversion of planned traffic to avoid conflict situations with unexpected flights Increased operational costs due to extra fuel consumptions as a result of modifications of descending paths in case deconflicting actions are required
Secondary effects	Increased flight crew workload due to action imposed by ATC to preserve safety Rescheduling of TMA incoming traffic in order to adapt this flow to the complexity of the current traffic Increased delays on affected flights and on all traffic that follows Use of backup technology with an associated decrease of system capacity and efficiency

Table 3 Description of a 1090 MHz jamming scenario

Scenario	Description
Intended attack	Reduction (or even elimination in the case of ADS-B exclusive environments) of surveillance capabilities by blocking the access to accurate information provided by the aircraft, leading to loss of multiple tracks
Primary impact	Lack of up-to-date flight information provided by onboard systems Need for backup surveillance systems (PSR) Null capability for identifying detected targets Partial loss of situational awareness
Secondary effects	Increased separation between flights Increased delays of affected flights and all traffic that follows Use of backup technology with an associated decrease of system capacity and efficiency In the case of ADS-B exclusive environments, emergency procedures need to be followed to ensure safety

B. 1090 Megahertz Jamming Scenario

A frequency of 1090 MHz is used indistinctly by the secondary surveillance radar transponder and by the ADS-B system. Although information is different in both cases, the sending of those packages is made by using the same signal carrier. Hence, interfering or jamming this signal deactivates, at the same time, the onboard capabilities of answering the SSR queries and broadcasting the current aircraft position, velocity, and/or any other relevant flight information.

Usually, a jammer consists of a transmitter that, with the aid of an omnidirectional antenna, emits in a specific frequency (1090 MHz in this case), interfering with CNS/ATM receivers. The jammer transmits a signal that is radiated in all directions with enough emission power to jam the 1090 MHz receivers within the area of influence.

This action can target a specific aircraft if the jammer is installed onboard or if a passenger carries out the jamming, but it also can target traffic approaching an airport if the jammer is located on the ground. The higher the jammer emission power, the greater the number of flights that can be affected.

A worst-case scenario is represented by an ADS-B- or WAM-exclusive environment, with radar redundancy in which the jammer affects all flights within the considered airspace; and it affects a significant number of tracks in the sector. In this case, the situation could become unmanageable due to the total loss of situational awareness (loss of multiple tracks). Only direct communications through control pilot datalink communications, or even through the vhf channel, could be used for identifying and managing the traffic, which would also increase the stress and workload for both ATCOs and flight crews. In environments with redundant surveillance systems (WAM, radar), the situation would be manageable in all cases because reverting to procedural operations would be possible. However, efficiency, and therefore operational costs, would be unavoidably affected. Table 3 summarizes the described attack scenario.

C. Cyberattack Scenario

Main cyberattacks are conceived as attacks that affect one or more functionalities or datasets managed by the SWIM infrastructure. However, there are other alternatives that can cause a big impact on standard operations and are considered as cyberattacks. The fact of extending IT solutions to the communication between the manufacturer, airline, maintenance centers, and suppliers opens the door to new cybervulnerability sources. The more critical the systems, the more critical it is to protect them. This is true in the case of malware uploads through the airplane assets distribution system (AIADS). The AIADS enables the distribution of valuable information that ranges from structural health monitoring data or traffic control data to loadable software. For example, an intended insider can upload certain software pieces through the dedicated aircraft

Table 4 Description of a cyberattack scenario

Scenario	Description
Intended attack	Upload of a malware application for generating a false alarm onboard during a flight Reduction of aircraft capabilities when airborne
Primary impact	Application of emergency procedures Route clearance of the target flight if the false alarm is related to a safety-critical system Diversion of surrounding traffic to facilitate emergency procedures Increased air traffic controller and flight crew workload
Secondary effects	Impact on airport operations Increased delays, both for flying and landed aircraft Increased maintenance costs due to the need of exhaustive analysis of the incident

network with the goal of corrupting any relevant data, denying the service of some onboard equipment, producing disfunctionalities that can result in false alarms or leak operational data to a third party. Table 4 summarizes the described attack scenario.

The situation might become unmanageable if the malware is installed in an entire airline fleet (or in a large number of its airplanes) and the execution is simultaneously coordinated. If just one flight is affected, there are emergency procedures in place that ensure the aircraft can land safely in a nearby aerodrome. However, if several aircraft of the same airline are suffering from the same false alarm when operating within the same airspace, the system could be overloaded due to the need of applying excessive emergency procedures to large number of flights.

VI. Conclusions

Future ATM systems will be netcentric operational scenarios, highly dependent on information technologies. The ATM technologies that are currently under development provide new functionalities and capabilities and, at the same time, may enable new vulnerabilities that can be exploited by intentional attackers. The new ATM paradigm, in particular, will rely on the capability of many stakeholders with different interests to access the right information at the right place and the right time (netcentricity). However, this information could be manipulated (e.g., corrupted or removed) and/or the access to it denied or interrupted, with the intention of disturbing standard operations. Additionally, advanced technologies or e-enabled aircraft are also subjected to intentional attacks with the goal of dismissing the efficiency of standard operations, and even disrupting some maintenance or operational functions. This paper has presented a study of some main vulnerabilities of the CNS/ATM system based on the release and exchange of information. Threats that can result from the potential access to the flow of information among stakeholders are also considered.

A description of the information domains that led to the development of the future system has been included. Understanding how the information is organized helps the process of identifying levels of information criticality, which in turn supports the analysis of main sources of vulnerability.

A review of vulnerabilities that will face the future ATM system focused on information accessibility has also been included. Based on that, three different attack scenarios have been described that consider the same operational environment.

For the purpose of designing effective countermeasures, a thorough study of any possible individual attack is needed. However, it is necessary to understand that any realistic attack will consist of several individual attacks for which the particular objectives are different and apparently disconnected but which share a greater goal through a sophisticated common plan.

Impact scalability has been identified as a relevant factor to analyze meticulously. Some individual attacks produce impacts that can be managed by the system more or less effectively. The problem arises when the number of such attacks, which individually do not produce a major system disruption, is so high that the system capacity is overwhelmed.

Multithreaded and scalability analyses support the concept of developing technologies to reduce the impact of any security incident, regardless of the type of derived impact. Any suspicious behavior, although initially negligible, needs to be solved because it can potentially be part of broader and more sophisticated attacks that are more difficult to manage once some security barriers are breached.

Recurrent assessment to identify threats and vulnerabilities is needed and should be the basis for air and ground security requirements. Threat information sharing, shared learning, and collaborative efforts will minimize risks and impacts. In this regard, this paper also suggests addressing the ATM information security from a broad approach that is not limited to the acquisition of security technologies but based on well-established security programs already implemented in other critical environments such as the financial or military domains.

Finally, ATM security and cybersecurity require new approaches (mainly cultural), as well as training and new tools for decision makers to make more informed decisions on information security management.

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